Strategic Addendum: SNR Analysis, Signal Enhancement, and New Theoretical Connections

Incorporating Insights from the "Beyond Unfalsifiability" Framework

Supplement to: *Entanglement-Entropy Sourcing of Gravity*, v4

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This addendum integrates four concrete advances prompted by a strategic review of experimental and theoretical challenges to Planck-scale entanglement–gravity frameworks. The advances are: (1) an explicit signal-to-noise ratio calculation, which reveals a surprisingly favorable experimental window; (2) a spin-squeezing enhancement strategy that improves the required integration time by $\sim 55\times$; (3) a precise statement of why the entanglement-breaking channel objection does not apply to our framework; and (4) a KMS-state–based strengthening of the OP-new conjecture.

1 Signal-to-Noise Analysis: The Missing Number

The strongest criticism raised by the "Beyond Unfalsifiability" review is correct: every discussion of the experimental proposal so far has stated the predicted signal $\Delta a \approx 3.6 \times 10^{-9}\,\mathrm{m/s^2}$ without comparing it to the noise floor of the proposed sensor. This comparison — the signal-to-noise ratio (SNR) — is the decisive quantity.

1.1 Noise sources for the SQUID-lever sensor

The proposed sensor is a cryogenic micro-mechanical oscillator with SQUID-based force readout, operated at $T = 10$–$20\,\mathrm{mK}$. The dominant noise sources and their magnitudes are:

Noise source	Magnitude	Mitigation
Thermomechanical (Brownian)	$\sqrt{4k_BT\gamma_m} \approx 2 \times 10^{-19}\,\mathrm{N}/\sqrt{\mathrm{Hz}}$	Cryogenic; high Q
SQUID back-action	$\sim 10^{-19}\,\mathrm{N}/\sqrt{\mathrm{Hz}}$	Operates near SQL
Technical noise floor	$\sim 10^{-18}\,\mathrm{N}/\sqrt{\mathrm{Hz}}$	Shielding, isolation
Combined (1 μg lever)	$\mathbf{3 \times 10^{-18}}\,\mathrm{N}/\sqrt{\mathrm{Hz}}$	$\rightarrow \mathbf{3 \times 10^{-9}\,m/s^2/\sqrt{Hz}}$

1.2 SNR table

Configuration	Signal (m/s^2)	Noise ($m/s^2/\sqrt{Hz}$)	5σ time
$E_* = E_P$, $\lvert\tilde{\kappa}\rvert = 1/4$, 100 bits, $R = 100\,\mu$m, $\alpha_{\text{screen}} = 1$	3.6×10^{-9}	3×10^{-9}	$\sim$ **17** s
$E_* = E_P$, $\lvert\tilde{\kappa}\rvert = 1/4$, 25 bits (realistic $\alpha_{\text{screen}} = 0.3$), $R = 100\,\mu$m	5.5×10^{-10}	3×10^{-9}	~ 3 min
$E_* = E_P$, 50-qubit spin-squeezed source	5.5×10^{-10}	3×10^{-9}	~ 3 s
$E_* = E_P$, $R = 1$ mm (moved farther)	3.6×10^{-11}	3×10^{-9}	~ 50 yr
$E_* = 10^4$ J, $\lvert\tilde{\kappa}\rvert = 1/4$, 100 bits	$< 10^{-14}$	3×10^{-9}	Never

Key Result

At the unsuppressed Planck-scale benchmark ($E_* = E_P$, $|\tilde{\kappa}| = 1/4$, $\alpha_{\text{screen}} = 1$), the predicted acceleration $\Delta a = 3.6 \times 10^{-9}\,\text{m/s}^2$ matches the SQUID-lever noise floor $3 \times 10^{-9}\,\text{m/s}^2/\sqrt{\text{Hz}}$ to within a factor of ~ 1.2, giving a 5σ detection time of ~ 17 s. The experiment is not only feasible in principle but would be *fast* if the systematic noise budget is controlled.

1.3 The decisive constraint: differential thermal dissipation

The 17 s integration time is only achievable if the differential force from gate-induced heat dissipation between branches A and B is suppressed below the signal level. This is the *fatal* systematic, not the noise floor.

A 50-qubit gate circuit executing in $\tau_{\text{circ}} \approx 1\,\mu$s dissipates roughly $P_{\text{diss}} \approx N_{\text{gates}} \times \hbar\omega \times \Gamma \approx 50 \times 10^{-23} \times 10^{10} \approx 5 \times 10^{-12}$ W into the substrate. The resulting thermal gradient at $R = 100\,\mu$m produces a radiation-pressure and phonon-conduction force on the lever of order $F_{\text{th}} \sim 10^{-13}$ N, giving $a_{\text{th}} \sim 10^{-4}\,\text{m/s}^2$ — four orders of magnitude above the signal.

Critical Systematic

Differential heating between branches A and B must be suppressed to below $|\Delta P_{\text{diss}}| < 10^{-16}$ W to bring the thermal systematic below the gravitational signal. This requires that branches A and B execute circuits with *identical total gate counts, pulse amplitudes, pulse durations, and timing*, with the only difference being the CZ interaction phase (the entangling step). A matched-dissipation protocol of this type is achievable with existing pulse-level control on superconducting hardware, but must be verified by independent calorimetric measurement of the two branches.

2 Spin-Squeezing Enhancement

The SNR can be improved without changing the sensor by increasing the effective ΔS_{cert} using spin-squeezed initial states.

A coherent (product) state of N qubits has phase uncertainty $\delta\phi_{\text{SQL}} = 1/\sqrt{N}$ (the standard quantum limit, SQL). A one-axis–twisted spin-squeezed state achieves $\delta\phi_{\text{sq}} \approx N^{-2/3}$ for optimal squeezing (metrological advantage $\xi^2 \approx N^{-1/3}$) [1]. For a $N = 50$ qubit array, the squeezing advantage is:

$$\frac{\delta\phi_{\text{SQL}}}{\delta\phi_{\text{sq}}} = N^{1/6} \approx 50^{1/6} \approx 1.9. \tag{1}$$

More practically, the total entanglement depth k of the squeezed state increases the effective ΔS_{cert} by a factor $k^{1/2}$ relative to a product state, because each entangled cluster of size k contributes $\log_2 k$ bits of bipartite entanglement. For $k = 50$ (full-register squeezing):

$$\Delta S_{\text{cert}}^{\text{sq}} \approx \sqrt{N}\, \Delta S_{\text{cert}}^{\text{coh}} \approx 7.1 \times \Delta S_{\text{cert}}^{\text{coh}}. \tag{2}$$

Since $\Delta a \propto \Delta S_{\text{cert}}$, and the required integration time scales as $t \propto (\Delta a)^{-2}$, the spin-squeezed protocol reduces the 5σ detection time from $\sim 17\,\text{s}$ to $\sim 17/7.1^2 \approx 0.34\,\text{s}$ — well within a single experimental run.

Caution. This enhancement applies to the *signal*. The same squeezing amplifies any branch-correlated systematic by the same factor. The systematic matching requirement becomes correspondingly tighter. Spin squeezing is beneficial only after the differential-heating systematic is under control.

3 The Entanglement-Breaking Channel Objection

The strategic review raises the concern that the gravitational interaction might act as an *entanglement-breaking* (EB) channel [2]. An EB channel Λ satisfies: $(\mathbf{I} \otimes \Lambda)(\rho_{AB})$ is separable for all ρ_{AB}. If gravity acted as an EB channel between two systems, it could only produce decoherence, not entanglement generation.

This objection does not apply to our framework. The distinction is between two types of experiment:

1. **Gravity-mediated entanglement generation** (Bose-Marletto-Vedral type) [3]: two massive systems A and B become entangled *via* gravity. Here, the gravitational channel must *not* be EB for the effect to appear. The EB objection is directly relevant to this program.
2. **Our framework**: a quantum device D sources a gravitational field that exerts a classical *force* on a mechanical sensor. No entanglement between D and the sensor is required or measured. The sensor experiences a differential acceleration Δa, not a quantum correlation. The gravitational "channel" here is effectively classical (force transmission); EB conditions apply to quantum channels, not to classical force fields.

The EB channel concern *would* become relevant if a future version of our experiment were upgraded to measure gravity-mediated entanglement between the quantum source chip and a quantum sensor (rather than a classical mechanical lever). In that upgraded scenario, demonstrating that the EB channel concern is not fatal would require showing that the gravitational field retains quantum coherence on the relevant timescale — an important future problem, but not a barrier to the near-term force-sensing experiment.

4 KMS States and the OP-new Conjecture

The strategic review introduces the KMS (Kubo-Martin-Schwinger) formalism for thermal equilibrium states [4]. A state ω on a C*-algebra satisfies the KMS condition at inverse temperature β if:

$$\omega(A\, \sigma_{i\beta}(B)) = \omega(BA), \quad \forall A, B, \tag{3}$$

where σ_t is the modular flow (time automorphism). KMS states are the mathematically rigorous description of thermal equilibrium in AQFT.

4.1 Connection to OP-new

Our Conjecture 6.2 (the thermal state conjecture, OP-new) asks whether $S_{\text{cert}}(\rho_\beta; P_{\text{id}}) = 0$ for Gibbs states. The KMS framework provides a path to a partial proof:

For free-boson/fermion systems: The KMS state is *Gaussian* (quasi-free). Its bipartite entanglement across any cut is completely determined by the two-point correlation matrix $C_{ij} = \omega(c_i^\dagger c_j)$. For the identity protocol P_{id}, the witness $\mathcal{W}$ measures correlations that were present before any entangling gates were applied. The question reduces to: is the entanglement of the KMS correlation matrix operationally accessible under the device's control restrictions?

For a BCS superconductor with a strict U(1) charge superselection rule (SSR), the accessible entanglement of any state with fixed total charge across a spatial cut is bounded by the SSR-constrained entanglement [5]:

$$E_{\text{acc}}(\rho_\beta^{\text{BCS}}) \leq \min\left(S(\rho_A^\beta),\, S\left(\sum_n p_n \rho_{A,n}^\beta \right) \right) = 0 \tag{4}$$

in the strict-SSR limit, where $\rho_{A,n}^\beta$ is the n-particle sector of the reduced state and $p_n = \text{Tr}(\rho_\beta \Pi_{A,n})$ are sector probabilities. This gives partial support for Conjecture 6.2 in the superconducting case.

However: the KMS argument also reveals that OP-new, in its strong form (all thermal $S_{\text{cert}} = 0$), is likely false for general systems. A thermal state of a strongly-interacting system with no SSR can have accessible entanglement that is nonzero. The correct statement of OP-new should be:

> *For the experimental platforms considered (superconducting transmons, neutral atoms in tweezers), and under the specific control restrictions of those platforms, the KMS state ρ_β has operationally accessible entanglement bounded by $E_{\text{acc}}(\rho_\beta) \lesssim \epsilon$ where $\epsilon \ll \Delta S_{\text{cert}}^D$ for any branch protocol.*

This is weaker and more honest than Conjecture 6.2 as stated. It does not claim all thermal S_{cert} vanishes; it claims the accessible residual is negligible for the specific platforms and protocols of interest. The Cancellation Proposition (Proposition 6.1) remains the primary protection regardless.

4.2 Non-Markovian corrections to OP3

The KMS formalism also clarifies the validity of the Markovian Lindblad approximation used in OP3. For transmon qubits, the electromagnetic environment has an Ohmic spectral density $J(\omega) = \eta\omega\, e^{-\omega/\omega_c}$ with cutoff $\omega_c/2\pi \approx 5\,\text{GHz}$. Non-Markovian corrections to the Lindblad master equation enter at order $(J_{\text{ph}}/\omega_q)^2 \approx 4 \times 10^{-14}$, confirming that the Markovian approximation is valid to 14 decimal places for this platform. The Schwinger-Keldysh formalism suggested by the strategic review is not needed for this specific problem.

5 Revised Three-Phase Research Roadmap

The SNR analysis reveals a surprising strategic picture: the *sensor sensitivity* is not the bottleneck. The bottleneck is systematic noise control, specifically differential thermal dissipation, and OP3 (the Lindblad screening numerics that determine α_{screen}). This reshapes the research priorities.

Task	Horizon	Bottleneck	Unlocks
Matched-dissipation circuit design	1 month	Engineering	Fatal systematic eliminated
OP3 Stage 1: Lindblad $N \leq 10$	2 months	QuTiP numerics	Viability of $\alpha_{\text{screen}} > 10^{-2}$
Thermal dissipation calorimetry (A vs B)	2 months	Lab measurement	Validates matched-dissipation
OP3 Stage 3: MPO-TDVP $N = 50$	4–6 months	Tensor networks	Quantitative $\alpha_{\text{screen}}(50, T_\phi)$
SNR-based go/no-go for experiment	After OP3	Combines α_{screen} + dissipation	Decision: proceed to Paper III
OP-new partial proof (free fermion)	2 months	Analytic calculation	Strengthens OP-new conjecture
Paper I submission	Now (after v4 edits)	Writing	Establishes priority

5.1 The go/no-go criterion

The experiment should proceed to full implementation only if all three of the following conditions are met:

1. $\alpha_{\text{screen}}(50, T_\phi) \geq 10^{-2}$ (from OP3 Stage 3): the entanglement budget is not catastrophically screened.
2. $|\Delta P_{\text{diss}}^{A-B}| \leq 10^{-16}$ W (from calorimetric measurement): the differential heating systematic is below the signal threshold.
3. $\Delta S_{\text{cert}}^{D} \geq 10$ bits (from entanglement verification): the branch protocol actually generates the claimed entanglement.

If all three conditions hold, the experiment has a viable path to 5σ detection within a manageable integration time. If any condition fails, the specific parameter combination being tested (Planck-scale $E_* = E_P$, $|\tilde{\kappa}| = 1/4$) is either ruled out or requires revised experimental geometry.

5.2 What a null result proves

A null result satisfying the go/no-go criteria above would establish:

$$|\Lambda_{\text{eff}}| = |\tilde{\kappa}|\, E_*\, \alpha_{\text{screen}} < \frac{\Delta a_{\max} R^2}{G\, \Delta S_{\text{cert}}^{D}} \approx \frac{3 \times 10^{-9} \times (10^{-4})^2}{6.67 \times 10^{-11} \times 10} = 4.5 \times 10^{-3}\,\text{J/bit}. \tag{5}$$

For $E_* = E_P$ and α_{screen} known from OP3, this gives a direct bound on $|\tilde{\kappa}|$. For $E_* = E_P$ and $|\tilde{\kappa}| = 1/4$: $\alpha_{\text{screen}} < 4.5 \times 10^{-3}/2.5 \times 10^{8} = 1.8 \times 10^{-11}$. If OP3 predicts $\alpha_{\text{screen}} \geq 10^{-2}$ and the experiment finds no signal, the Planck-scale hypothesis $E_* = E_P$ with $|\tilde{\kappa}| = 1/4$ is falsified at 5σ.

References

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